

also offers a highly capable and scalable open source cloud management stack. You can see it at work on buzzing public clouds like Cloudwatt, DreamCompute, eNocloud, HP, Rackspace and Ulticloud. Or, if you are a developer, you can explore its dashboard and features by installing Devstack (<http://devstack.org/>) on your PC!

OW2, on the other hand, claims to offer something totally different from these—in fact, the developers claim to offer a complementary rather than competing service. As a PaaS provider, it helps companies manage the heterogeneity in their social media applications, data, resources and services comprising cloud-oriented information systems. OW2 offers middleware that uses the resources provided by different technologies of different companies.

Another interesting project in this space is OpenCloudWare, which is working to build a self-service platform to manage the lifecycle of virtualised multi-tier applications such as JavaEE—OSGi. The components will include modelling, developing and building images, a multi-IaaS compliant PaaS platform, orchestration, performance testing, self-management features like elasticity, green IT optimisation and provisioning.

Speaking of clouds, Dushyanth Harinath, senior director, System Operations, BigRock, brings an interesting project to our notice: “Docker (www.docker.io) is an open source project that allows developers and operations staff to quickly deploy their applications as lightweight, portable and self-sufficient containers. It has kicked up such excitement and resulted in so many contributions from the open source community that it now supports various public and private clouds like AWS, OpenStack, Google Compute Engine, etc. It also powers PaaS/SaaS platforms such as Domino (a PaaS for data science), Blitline (a SaaS service for image processing), OpDemand (Opensource PaaS) and Dokku (a mini-heroku open source PaaS). Docker’s impact on development and operations teams is immense due to its ‘build once, run anywhere’ motto. With Docker, the full lifecycle of an application, i.e., right from the development systems, to staging and production, can be streamlined consistently. What this means is, a development team can run its applications on an isolated Docker container on Linux machines, which then can be deployed in production on the supported public or private clouds, seamlessly. This results in a very efficient and repeatable application delivery process.”

Building the Internet of Things

Trend: We are on the verge of the Internet of Things (IoT) era, in which everyday objects including refrigerators, washing machines, gates and chairs will be uniquely identifiable and connected to the Internet, enabling an unprecedented amount of authorised control, customisation and automation. The foundations for this are being built rapidly—communication technologies, protocols, embedded systems and software. Almost every IT and consumer

electronics major is interested in this space.

Perhaps because of the inherent diversity in a system such as the IoT, we find open hardware, software and protocols to be quite popular in this arena. Definitely, objects that are to be connected need to be based on an open architecture, if they are to be utilised well by myriad applications and tools.

Trend++: Hardware, software, platform or protocol, whatever you need to build the IoT is available as open source. Contiki, one of the most popular operating systems for IoT products, is completely open source. It is cutting-edge, and has been developed with contributions from some of the most well-known industry players. It provides powerful low-power Internet communication, supports IPv6 and IPv4, and the newest low-power wireless standards including 6LoWPAN, RPL and CoAP. By smartly implementing sleepy routers and a radio duty cycling mechanism, Contiki enables relay nodes (wireless routers) to be battery-operated. Thingsquare, a company founded in 2012 by Contiki’s creator Adam Dunkels, offers open source platforms that help add wireless mesh networking and Internet connectivity to any and every product. It is being used increasingly in several interesting products such as Tado, the smart thermostat.

ThingSpeak, another upcoming project, helps to send and receive information from gadgets, such as blood sugar readings from a glucose meter or usage information from smart air-conditioners, using simple HTTP requests. Once the data is in ThingSpeak, you can build applications that use that data for various purposes.

PaaS offerings for IoT products are also popping up. Nimbits is an open source IoT PaaS that helps develop software and hardware solutions, which seamlessly connect to the cloud and to each other. At the core is the Nimbits Server, which provides REST Web services for logging and retrieving time and geo-stamped data, such as the readings from sensors. The Nimbits server is versatile enough to run on large cloud platforms like the Google App Engine to the smallest Raspberry Pi device. Xively, another popular PaaS, offers an IoT platform and open source libraries, along with services and consultancy.

Another interesting example, the IoT Toolkit, includes a smart object API as well as a gateway-as-a-service program. Binary images for a RaspberryPi running Ubuntu are available for instant access to the tools. Gateways with open designs, such as the Freescale/Oracle OneBox, are also available. The Open Home Gateway Forum offers an open software design for the gateway.

Coming to the hardware, once again, open source is giving the IoT a big push. If you want everything to be connected, obviously the movement has to be powered not just by companies but individuals too. Do you know how to automate your old fan? You could find out, if you experiment with open hardware like Arduino and Raspberry Pi! Apart from the well-known names, there are also some specific boards for the IoT such as Pinoccio, a wireless, Web-ready